# OBJECT ORIENTED PROGRAMMING

**ACTIVITY #2 - Operators**

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## General Instructions:

1. Create a Java program based on the questions.
2. Follow the exact required output shown in the question.
3. You must use the specific operator required in each number.
4. Do NOT hardcode the final answers (e.g., System.out.println(86); without computation is not allowed).
5. Show proper use of variables and correct operator usage.
6. Follow proper Java syntax, indentation, and naming conventions.
7. Use System.out.println() for displaying output.
8. Screenshot the code and the output.
9. Compile and run your program to verify correctness before submission.
10. Submission Format: Section_LastName_FirstName_OperatorsActivity#2.pdf

## Part 1

Create **three (3) separate Java programs** following the required operators:

1. A program that uses **bitwise operators**.
2. A program that uses the **ternary operator (? :).**
3. A program that uses **assignment operators (+=, -=, *=, /=)**.

Each program must **produce the exact output required in the question** and must compute the result using the specified operator.

## Bitwise Operators

```
bitwiseoperator.java > ...
public class bitwiseoperator {
    Run | Debug
    public static void main(String[] args) {
        int a = 15;
        int b = 60;

        System.out.println("Bitwise AND: " + (a & b));
        System.out.println("Bitwise OR: " + (a | b));
        System.out.println("Bitwise XOR: " + (a ^ b));
    }
}
```

## Output

```
ceptionMessages' '-cp' 'C:\Users\ACER\AppData\R
c1571c3f\bin' 'bitwiseoperator'
Bitwise AND: 12
Bitwise OR: 63
Bitwise XOR: 51
```

## Ternary Operators

```
public class ternaryoperator {
    Run | Debug
    public static void main(String[] args) {

        int age = 18;
        int age2 = 14;

        System.out.println("Your age is " + age + " and you " + (age >= 18 ? "Get In" : "Get Out"));
        System.out.println("Your age is " + age2 + " and you " + (age2 >= 18 ? "Get In" : "Get Out"));
    }
}
```

## Output

```
ceptionMessages' '-cp' 'C:\Users\ACER\
c1571c3f\bin' 'ternaryoperator'
Your age is 18 and you Get In
Your age is 14 and you Get Out
```

## Assignment Operators

```java
public class assignmentoperator {
    Run | Debug
    public static void main(String[] args) {

        int a = 10;

        System.out.println("Initial Number: " + a);

        a += 1;
        System.out.println("Number when a += 1: " + a);
        a -= 5;
        System.out.println("Number when a -= 5: " + a);
        a *= 7;
        System.out.println("Number when a *= 7: " + a);
        a /= 2;
        System.out.println("Number when a /= 2: " + a);
        a %= 4;
        System.out.println("Number when a %= 4: " + a);
    }
}
```

## Output

```
c1571c3f\bin' 'assignmentoperator'
Initial Number: 10
Number when a += 1: 11
Number when a -= 5: 6
Number when a *= 7: 42
Number when a /= 2: 21
Number when a %= 4: 1
```

## Part 2

Create one Java program that demonstrates the use of multiple operators.

Requirements:

1. Use at least five (5) different operators from the following categories:

- Arithmetic operators
- Relational operators
- Logical operators
- Unary operators
- Assignment operators
- Bitwise operators.

2. The program must produce exactly six (6) outputs only.
3. At least one output must contain a tricky increment or decrement expression, such as:

- a++ + ++a
- --b + b++
- ++x + x++

4. Do not reuse variable names, values, or logic patterns from Part I.
5. The outputs must include:

- At least two (2) boolean results
- At least four (4) numeric results

## Multiple Operators Demo

```java
public class MultipleOperatorsDemo {

    Run | Debug
    public static void main(String[] args) {

        int x = 7;
        int y = 12;
        int z = 3;

        // 1. Arithmetic operator
        int result1 = x * y + z;
        System.out.println("Arithmetic operator: " + result1);

        // 2. Assignment operator
        int result2 = y;
        result2 += 8;
        System.out.println("Assignment operator: " + result2);

        // 3. Bitwise operator
        int result3 = x & y;
        System.out.println("Bitwise operator: " + result3);

        // 4. Relational operator
        boolean result4 = x > y;
        System.out.println("Relational operator: " + result4);

        // 5. Logical operator
        boolean result5 = (x < y) && (z > 1);
        System.out.println("Logical operator: " + result5);

        // 6. Tricky increment expression
        int result6 = x++ + ++x;
        System.out.println("Tricky Operator: " + result6);

    }
}
```

## Output

```
s\ACER\AppData\Roaming\Antigrav
Arithmetic operator: 87
Assignment operator: 20
Bitwise operator: 4
Relational operator: false
Logical operator: true
Tricky Operator: 16
```